

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What is size reduction? With construction & working, explain the working of roll crusher.

Q.24 Define sedimentation. Explain different sedimentation process and also write about rate of sedimentation.

Q.25 Write short note on any two of following:

- a) Screening
- b) Cake filtration
- c) Characteristics of filter media

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Roll No.

2nd Sem / Chemical, Chem P & P

Subject : Mechanical Operations

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In sieve analysis, particles are classified based on:

- a) Shape
- b) Size
- c) Density
- d) Porosity

Q.2 If the sphericity of a particle is 1, the particle is:

- a) Cube
- b) Sphere
- c) Cylinder
- d) Irregular

Q.3 Which of the following is a fine grinder?

- a) Jaw crusher
- b) Smooth roll crusher
- c) Ball mill
- d) Grizzly

Q.4 Mesh number indicates:

- a) Opening size of screen

- b) Number of openings per linear inch
- c) Thickness of wire mesh
- d) None

Q.5 Plate and Frame filter press is a :

- a) Clarifying filter
- b) Cake filter
- c) Sedimentation equipment
- d) Centrifugal Separator

Q.6 Mixing is carried out to:

- a) Increase viscosity
- b) Achieve uniform composition
- c) Decrease surface tension
- d) None

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 State Rittinger's law in words.

Q.8 Name the one method commonly used for particle size analysis in industries.

Q.9 Write one advantage of vibrating screens.

Q.10 Name one gravity classifier.

Q.11 Name example of one type of filtration process.

Q.12 Name one equipment used for liquid-liquid mixing.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the purpose of size analysis in chemical industries.

Q.14 Differentiate between average particle size and number of particles in a mixture.

Q.15 Explain about centrifugal filter.

Q.16 What is a Fluid Energy Mill? Mention its use.

Q.17 Explain Tyler standard screen series (U.S sieve series)

Q.18 Explain the principle and working of a vibrating screen.

Q.19 Explain the working of Rotary Drum Filter.

Q.20 Differentiate between cake filtration and deep bed filtration.

Q.21 Explain the working of a Double Arm Kneader.

Q.22 Difference between Impeller and propeller in mixing.